

Polynomial Regression

ML Assignment 1 | Tanmay Dixit | BT2024016

Data and objective

Separate polynomial regression models predict the Net Power Score (var1) and Thermal Anomaly Score (var2). Each supplied training file has 1,000 labelled rows. Each test file has 1,000 rows without target labels. All six inputs are used for var1 and all three for var2. The four files contain no missing values. Original test-row order is preserved.

Polynomial models

Each term has a total degree equal to the sum of its exponents. For example, $x_1^2 x_2$ has total degree 3. The search includes powers and interactions, with degree capped at 10 for var1 and 20 for var2. Monomial and Legendre bases are compared. Both represent polynomials. A fixed average of polynomial fits is also a polynomial, with degree no greater than the largest component degree. No tree, kernel with non-polynomial terms, or neural network is used.

Model selection

A seed-42 split gives 800 development rows and 200 holdout rows per problem. Broad searches on the development rows compare ordinary least squares, Ridge, Lasso, relaxed Lasso, and automatic relevance determination (ARD) regression. ARD learns a separate shrinkage level for each coefficient. Expanded columns are standardized using only the fitting fold. Sparse term selection and any coefficient refit also stay inside that fold.

The initial expanded Ridge search covers all-feature degrees 1-6 for var1 and 1-12 for var2, plus reduced-feature controls. Further sparse searches test var1 degrees 4, 5, 6, 8, 10 and var2 degrees 6, 8, 10, 12, 16, 20, followed by a finer penalty grid near the leading degrees. Ridge basis and degree-penalty searches extend to degree 8 and 20, respectively. The complete grids and attempted numerical fits are recorded with the code.

Nine individual finalists and two fixed equal-weight averages are compared per problem. Each is evaluated with three five-fold partitions of the same 800 development rows (seeds 42, 137, 2026). The lowest mean MSE over these 15 folds determines the final choice. These are model-selection scores, not an unbiased final test estimate. Repeated folds overlap.

Validation and final fitting

The chosen settings are fixed before the final holdout evaluation. The 200-row holdout was examined during preliminary experiments and excluded from subsequent tuning, so it is a reused validation set. After recording MSE and R-squared, the fixed models are refitted on all 1,000 labelled rows to create the submission predictions.

MSE is the mean squared prediction error; lower is better. R-squared is $1 - \text{SSE}/\text{SST}$; closer to 1 is better. Hidden-test scores cannot be computed because test targets were not supplied.

var1: Steam turbine optimization

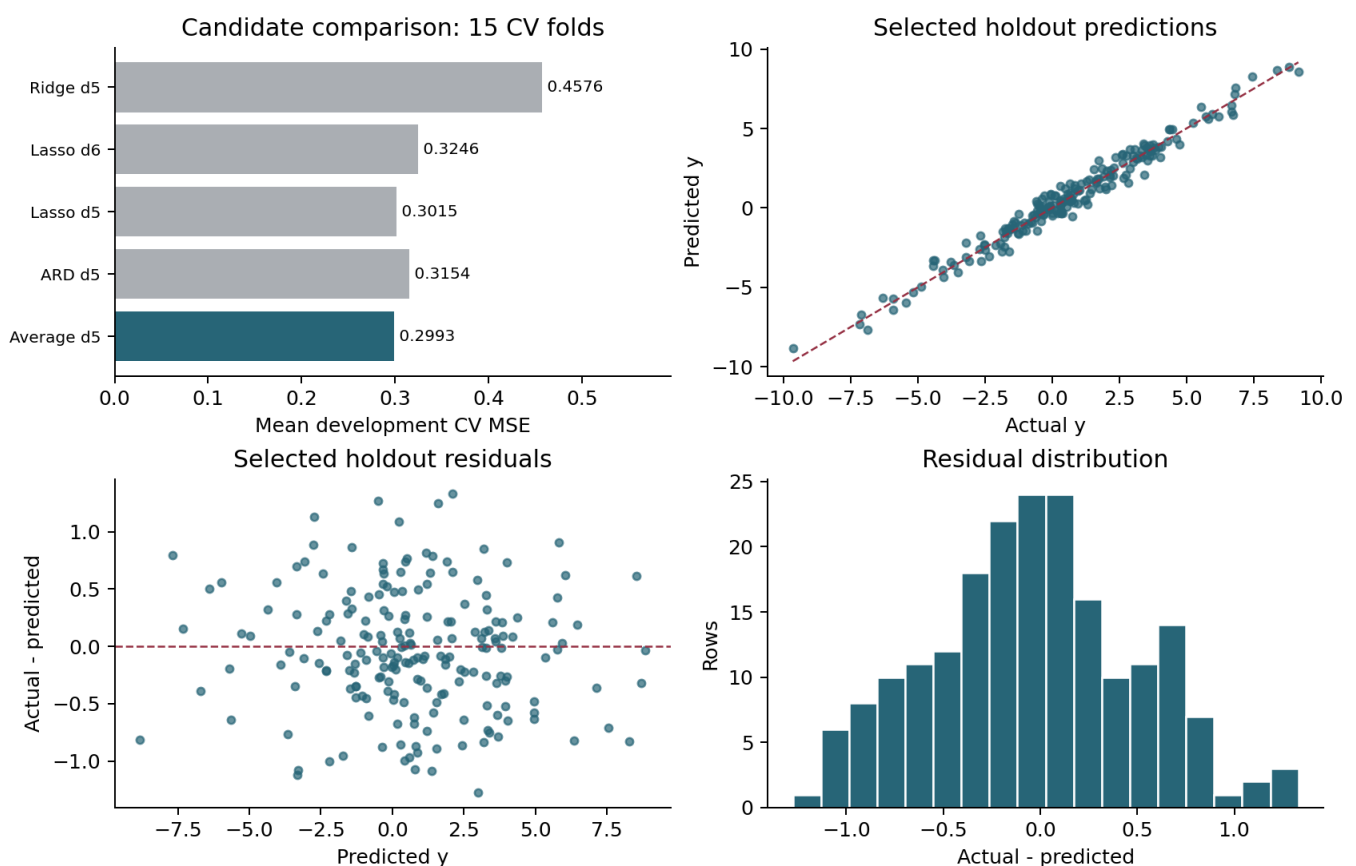
Selected model

Selected degree: 5. The model averages two degree-5 monomial fits with equal weights: Lasso (alpha = 0.008) and ARD (maximum 500 iterations, tolerance 0.0001). Both use all six features and start from 461 polynomial terms, excluding the intercept. Coefficient shrinkage limits the influence of unsupported powers and interactions.

Measured results

Repeated development CV MSE: 0.299307. Individual fold MSE standard deviation: 0.027033. This dispersion is descriptive; it is not a confidence interval.

Selected-model holdout MSE: 0.278873. Holdout R-squared: 0.971026. The chart compares representative candidates on the same 15 development folds. The full comparison covers 11 finalists per problem.



Selection rationale and final predictions

Degree-5 Lasso had lower repeated CV MSE than the tested degree-6 Lasso and dense degree-5 Ridge finalists. Averaging the degree-5 Lasso and ARD fits gave the lowest mean CV MSE among the finalists. Its advantage over Lasso alone is small and does not establish a statistically significant difference. Test inputs are more often near the feature boundaries than training inputs, so random-split performance may not fully represent hidden-test performance.

Final training uses all 1,000 labelled rows. BT2024016_pred_var1.csv contains 1,000 finite predictions under the single header y. The saved JSON model supports inference without retraining.

var2: Thermal reservoir mapping

Selected model

Selected degree: 12. The model equally averages a degree-12 Legendre Ridge fit ($\alpha = 0.01$) and a degree-9 monomial relaxed-Lasso fit ($\alpha = 0.001$). For Ridge, a standardized term of total degree k is divided by k^2 before fitting, so high-degree standardized coefficients receive a stronger penalty. The relaxed-Lasso fit averages its Lasso coefficients with a Ridge refit ($\alpha = 0.01$) on selected terms. The two expansions contain 454 and 219 terms, excluding intercepts.

Measured results

Repeated development CV MSE: 0.278119. Individual fold MSE standard deviation: 0.040501. This dispersion is descriptive; it is not a confidence interval.

Selected-model holdout MSE: 0.274358. Holdout R-squared: 0.995233. The chart compares representative candidates on the same 15 development folds. The full comparison covers 11 finalists per problem.



Selection rationale and final predictions

Degree-9 relaxed Lasso and degree-12 Ridge were competitive individual candidates. Their fixed average had lower mean repeated CV MSE than either component and was the best of the tested finalists. The resulting polynomial has maximum total degree 12. Hidden-test performance remains unknown.

Final training uses all 1,000 labelled rows. BT2024016_pred_var2.csv contains 1,000 finite predictions under the single header y. The saved JSON model supports inference without retraining.